deviceQualityDataManagement

Service Description

Service ID: *"deviceQualityDataManagement"*

Abstract
This document describes the `deviceQualityDataManagement` service, version 1.0.0, produced by the DeviceQoSEvaluator support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `deviceQualityDataManagement` service, which provides operator access to the stored device statistics and rebuilding of the device-system measurement map. It is produced by the DeviceQoSEvaluator system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interfaces are specified in the HTTP/TLS/JSON IDD [2] and the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

New in Arrowhead 5.

2.2 How This Service Is Meant to Be Used

Operators query aggregated statistics per metric group and period, and call `reload` after devices or systems changed in the ServiceRegistry.

2.3 Important Delimitations and Dependencies

Access follows the management policy of the DeviceQoSEvaluator.

3 Service Interface

This section describes the interfaces to the `deviceQualityDataManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

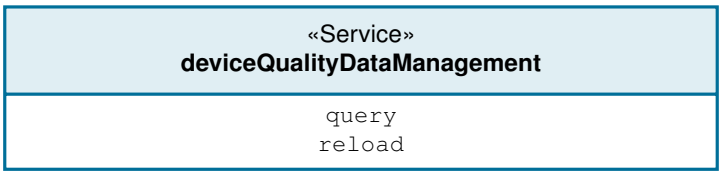


Figure 1: SysML-style block of the `deviceQualityDataManagement` service and its operations.

The following interface operations are available.

3.1 operation query (QoSDeviceStatQueryRequest) : QoSDeviceStatQueryResponse

Returns a page of aggregated statistic records.

3.2 operation reload () : String

Rebuilds the device-system measurement map from the ServiceRegistry.

4 Information Model

Here, all data objects that can be part of the `deviceQualityDataManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.5, which are used to represent things like hashes and identifiers.

4.1 struct Page

Paging and sorting; all fields optional.

Field	Type	Description
page	Number	Zero-based page number.
size	Number	Number of records per page (at most `max.page.size`, default 1000).
direction	String	Sort direction, `ASC` or `DESC`.
sortField	String	Field to sort by.

4.2 struct QoSDeviceStatQueryRequest

Paged filter for device statistics.

Field	Type	Description
pagination	Page	Paging and sorting of the result.
metricGroup	String	Metric group: `RTT`, `CPU_TOTAL_LOAD`, `MEMORY_USED`, `NETWORK_EGRESS_LOAD` or `NETWORK_INGRESS_LOAD`.
from	DateTime	Lower bound of the entry time.
to	DateTime	Upper bound of the entry time.
aggregation	List<String>	Statistics to return (minimum, maximum, mean, median, current).
systemNames	List<Name>	Systems whose devices are returned.

4.3 struct QoSDeviceStatQueryResponse

Device statistic records.

Field	Type	Description
records	List<QoSDeviceStatRecord>	Statistic records of the page.
count	Number	Total number of matching records (all pages).

4.4 struct QoSDeviceStatRecord

Aggregated measurements of one device in one period.

Field	Type	Description
metricGroup	String	Metric group: `RTT`, `CPU_TOTAL_LOAD`, `MEMORY_USED`, `NETWORK_EGRESS_LOAD` or `NETWORK_INGRESS_LOAD`.
id	Number	Identifier of the record.
timestamp	String	End of the aggregation period.
uuid	String	Identifier of the measured device.
minimum	String	Minimum value in the period.
maximum	String	Maximum value in the period.
mean	String	Mean value in the period.
median	String	Median value in the period.
current	String	Last measured value.
systems	List<String>	Systems hosted by the device.

4.5 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
List	An *array* of a known number of items, each having type A.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.

5 References

[1] J. Delsing, “DeviceQoSEvaluator – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-DeviceQoSEvaluator-java-spring/documentation/v5.3.0/DeviceQoSEvaluator-v5.3.0-SysD.pdf.

[2] ——, “deviceQualityDataManagement HTTP/TLS/JSON – Interface Design Description (DeviceQoSEvaluator),” Eclipse Arrowhead, IDD 5.3.0, 2026, deviceQoSEvaluator-v5.3.0-deviceQualityDataManagement-HTTP-IDD.pdf.

[3] ——, “deviceQualityDataManagement MQTT/TLS/JSON – Interface Design Description (DeviceQoSEvaluator),” Eclipse Arrowhead, IDD 5.3.0, 2026, deviceQoSEvaluator-v5.3.0-deviceQualityDataManagement-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	